

COURSE DESCRIPTION			
Program	Instrumentation Engineering		
Course Title	Basic Instrumentation Lab		
Course Code	2089	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

This course aims to provide hands-on experience in transducers and measurement systems, enabling students to apply instrumentation concepts through practical experiments.

Course Objectives

1	To infer the static characteristics, accuracy, precision, sensitivity, and errors of temperature measuring instruments.
2	To develop different types of bridge circuits for measurement of electrical parameters in instrumentation.
3	To demonstrate the characteristics of commonly used resistive transducers in instrumentation.
4	To apply the operation of Inductive, capacitive, magnetic, and optical transducers.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Fundamentals of Electronic instruments, passive and active components	1041	Elementary Concepts of Electronics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Determine the static characteristics and measurement errors of temperature measuring instruments using standard comparison methods.	Apply
CO2	Apply bridge circuits to determine unknown resistance, inductance, and capacitance.	Apply
CO3	Plot and interpret the characteristics of resistive transducers such as potentiometer, flex transducer and strain gauge.	Apply
CO4	Measure and analyse the performance of inductive, capacitive, magnetic, and photoelectric transducers used in instrumentation systems.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	3	2	-	-	-	-	-	2
CO2	3	3	2	3	3	-	-	-	-	-	2
CO3	3	2	2	3	2	-	-	-	-	-	2
CO4	3	3	2	3	3	2	-	-	-	-	2
Total	12	10	6	12	10	2	-	-	-	-	8
Correlation Level	3	2.5	2	3	2.5	2	-	-	-	-	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Static characteristics of an Instrument	To determine the accuracy, precision, and sensitivity of instruments such as bimetallic thermometer, vapour pressure thermometer, and mercury-in-steel thermometer.	CO1	Apply	3
2	Error of an Instrument	To determine the measurement error of an instrument such as bimetallic thermometer, vapour pressure thermometer, and mercury-in-steel thermometer by comparing with a standard instrument.	CO1	Apply	3
Module II					
3	Wheatstone's bridge.	To determine the unknown resistance using Wheatstone's bridge.	CO2	Apply	3
4	Hays bridge.	To determine the unknown inductance using Hays bridge.	CO2	Apply	3
5	Schering bridge	To determine the unknown capacitance using Schering bridge.	CO2	Apply	3
	Series Exam - I		CO2	Apply	3
Module III					
6	Potentiometer characteristics	To plot the characteristics of potentiometer	CO3	Apply	3
7	Flex transducer characteristics	To plot the characteristics of flex transducer	CO3	Apply	3
8	Strain gauge characteristics	To plot the characteristics of strain gauge	CO3	Apply	3
Module IV					
9	Level measurement using capacitive transducer	To measure level using Capacitive transducer	CO4	Apply	3
10	LVDT characteristics	To plot the characteristics of LVDT	CO4	Apply	3
11	Hall effect transducer	To find the magnetic polarity using hall effect transducer	CO4	Apply	3
12	Photoelectric transducers	To plot the characteristics of photodiode / phototransistor/ LDR	CO4	Apply	3
	Series Exam - II		CO4	Apply	3
	Open ended Experiment		CO4	Apply	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
Integrate multiple concepts from the syllabus into a combined application.
Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Read the procedure and precautions for experiments on static characteristics of measuring instruments.	1.5

2	Study methods to calculate accuracy, precision, sensitivity, and error from experimental readings.	1.5
3	Learn the importance of standard instruments in error measurement and calibration	1.5
4	Practice plotting graphs and writing results for instrument characteristic experiments.	1.5
Module II		
5	Study the circuit diagram and balance condition of Wheatstone bridge.	1.5
6	Read the working principle and applications of Hay's bridge	1.5
7	Study the working principle of Schering bridge and its use in capacitance measurement	1.5
8	Practice numerical problems related to resistance, inductance, and capacitance measurement using bridges.	1.5
Module III		
9	Read the experimental procedure for plotting characteristics of resistive transducers.	1.5
10	Learn graph plotting techniques for LDR experiment.	1.5
11	Study strain gauge mounting, Wheatstone bridge connection, and precautions.	1.5
12	Practice interpretation of characteristic curves obtained from transducer experiments.	1.5
Module IV		
13	Study the procedure and applications of capacitive level measurement applications	1.5
14	Read the working principle and experimental steps for LVDT and Hall effect transducers.	1.5
15	Study the procedure and applications of photodiode and phototransistor experiments.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Electrical and Electronic Measurements and Instrumentation	A. K. Sawhney	Dhanpat Rai & Co
2	Instrumentation and Measurement Systems	Raghuwanshi & Mittal	McGraw-Hill Education

REFERENCE			
Sl. No.	Title	Author	Publication
1	Handbook of Modern Sensors: Physics, Designs, and Applications	Jacob Fraden	Springer
2	Sensors and Transducers	D. Patranabis	Prentice Hall of India

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/108/108/108108051	I
2	https://www.next.gr/tutorials/sensors-and-transducers	I, III & IV
3	https://ieee-sensors.org/education-resources	I, III & IV
4	https://www.allaboutcircuits.com/textbook/direct-current/chpt-9/wheatstone-bridge	II
5	https://www.electronics-tutorials.ws	III & IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Bimetallic thermometer	Range: 0-200 °C	10
2	Vapour pressure thermometer	Range: 0-200 °C	10
3	Mercury in steel thermometer	Range: 0-200 °C	10
4	Standard thermometer	Mercury / Digital, calibrated	5
5	Wheatstone bridge kit	DC bridge with galvanometer	10
6	Hay's bridge kit	AC bridge with standard inductors	10

7	Schering bridge kit	AC bridge for capacitance measurement	10
8	Potentiometer transducer	Linear, 1-10 k Ω	10
9	Flex transducer	2.2 inch / 4.5 inch (commonly used)	10
10	LDR	CdS type	10
11	Strain gauge	Foil type, 120 Ω	10
12	Strain gauge trainer kit	With Wheatstone bridge & indicator	5
13	Capacitive level measurement setup	Tank with capacitive probe	5
14	LVDT trainer kit	± 5 mm range with signal conditioner	10
15	Hall effect sensor module	Linear Hall IC with magnet	10
16	Photodiode	Silicon type	10
17	Phototransistor	NPN type	10
18	Regulated DC power supply	0-30 V, dual output	10
19	Digital multimeter	3½ digit	10
20	CRO	20 MHz	2
21	Connecting leads & breadboards	Standard lab type	As required

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10

Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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